



CASE STUDY • GAME-X

From Browser to Build.

A custom PC Configurator for **Game-X** — turning complex hardware choices into confident, compatible purchases.



5,200+
Active SKUs



3
Custom Modules
Built



8wk
Delivery
Timeline



Compatibility Engine



Real-time Validation



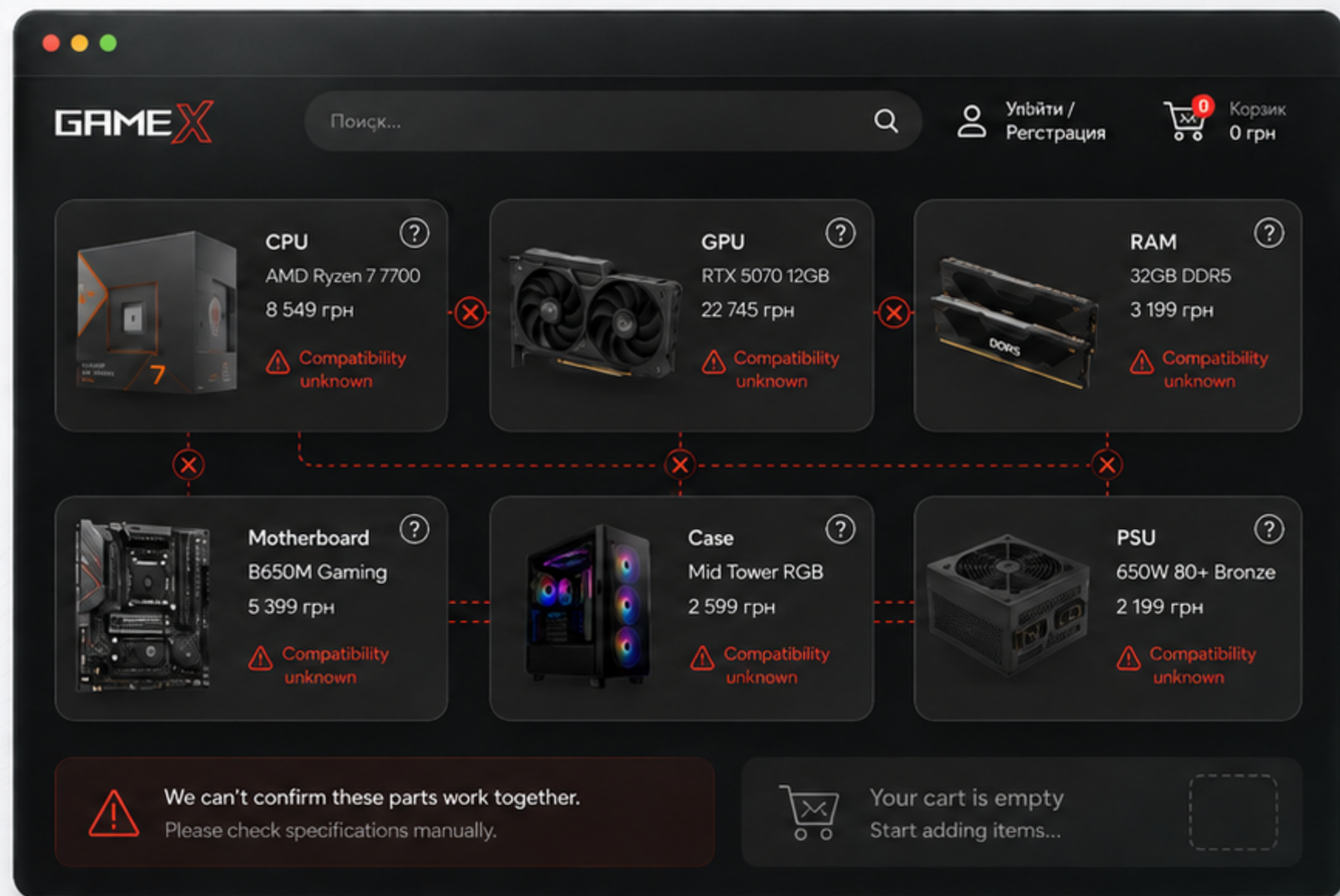
Smart Cart



01 / Challenge

Customers were lost. Carts were abandoned.

Game-X had 5,200+ SKUs across CPUs, GPUs, RAM, motherboards, cases, and PSUs — but no guidance. Buyers couldn't confidently pick compatible parts. Support was overwhelmed. Conversion suffered.



No compatibility filtering — customers ordered parts that didn't fit their build



No guided flow — product pages existed in isolation, no cross-sell logic



No configurator — competitors had one, Game-X didn't

02 / Solution

Three custom modules. One seamless engine.

GAMEX PC Configurator

Compatibility: All good | 350W+ Est. Power | 6 / 6 Steps Done

Step	Component	Status
1	CPU AMD Ryzen 7 7700	✓
2	Motherboard B650M Gaming	✓
3	RAM 32 GB DDR5	✓
4	GPU AMD Radeon RX 7800 XT 16GB	✓
5	PSU 650W 80+ Bronze	✓
6	Case Mid Tower RGB	✓

Compatibility Status
All components are compatible.

- ✓ Socket AM5 supported
- ✓ DDR5 memory supported
- ✓ mATX form factor compatible
- ✓ Estimated power usage: 352W

[View Details](#)

Validation in progress
Running extended compatibility checks... 78%
Checking GPU clearance and PSU connectors

Build Summary

Category	Value
Components	6 / 6
Estimated Power	352W
Compatibility	✓ All good

Total: **28 500 ₹**

[Add to Cart](#)



PC Configurator

Step-by-step builder, category-locked until previous step complete



Compatibility Engine

Real-time validation: socket, DDR gen, form factor, wattage



Smart Cart

Saves full builds, warns if a swap breaks compatibility before checkout

A guided buying flow that turns complex hardware selection into a confident purchase journey.

A builder that thinks like a technician.

- ✓ 6-step flow: CPU → Motherboard → RAM → GPU → PSU → Case
- ✓ Each step auto-filters next step's options by compatibility
- ✓ Live price counter updates as parts are added
- ✓ "Why is this locked?" tooltip explains issues in plain language
- ✓ Mobile-responsive

GAMEX

Search parts, models, brands...

Sign in / Register

Cart 2 items

Compatibility All good

Est. Power 352W

Slots Used 3 / 6

1 CPU

2 Motherboard

3 RAM

4 GPU

5 PSU

6 Case

AMD Ryzen 7 7700

B650M Gaming

32 GB DDR5

Select your GPU

Filtered by compatibility with your current build.



AMD Radeon RX 7800 XT
16GB
12 999 ₺
✓ Compatible



NVIDIA GeForce RTX 4070
12GB
16 499 ₺
✓ Compatible



AMD Radeon RX 6600
8GB
7 499 ₺
✓ Compatible



NVIDIA GeForce RTX 4090
24GB
52 999 ₺
⚠ Not compatible
Exceeds recommended power limit

Your Build

CPU
AMD Ryzen 7 7700

Motherboard
B650M Gaming

RAM
32 GB DDR5

4 GPU
Not selected

5 PSU
Locked

6 Case
Locked

Why is this locked?

Choose a GPU first to unlock compatible PSU and case options.



3 components selected
✓ All good! Everything is compatible.

Est. Power
352W

Total
28 500 ₺

Complete Build →

5,200 SKUs. Zero wrong combos.

The engine cross-references a relational compatibility matrix. Rules cover:

- 1. CPU socket ↔ Motherboard socket
- 2. RAM type ↔ Motherboard slot (DDR4/DDR5)
- 3. GPU power draw ↔ PSU wattage
- 4. Case form factor ↔ Motherboard size

Compatibility Engine

All rules validated

Selected Build

CPU

AMD Ryzen 7 7700

AM5

✓

Motherboard

B650M Gaming

AM5 • DDR5 • mATX

✓

RAM

32GB DDR5

6000 MHz

✓

GPU

NVIDIA GeForce RTX 5070

250W TDP

✓

PSU

750W 80+ Gold

750W available

✓

Case

ATX Mid Tower

Supports ATX / mATX

✓

Compatibility Matrix (Relational Rules)

✓ CPU socket ↔ Motherboard socket

AMD AM5 CPU

✓

AM5 Motherboard

✓ RAM type ↔ Motherboard slot

DDR5 RAM

✓

DDR5 Slots Motherboard

✓ GPU power draw ↔ PSU wattage

250W GPU TDP

✓

750W PSU Capacity

✓ Case form factor ↔ Motherboard size

ATX Mid Tower Case

✓

mATX Motherboard

✓ 4 / 4 layers validated

Build is 100% compatible

⌚ Real-time engine

Live rule evaluation

⦿ 0 conflicts

0 issues detected

4 Compatibility Layers

Real-time validation

0 invalid builds shipped

Built to convert. Proved it fast.

 **+34%**

Configurator-assisted
checkout conversion vs.
standard product page

 **-60%**

Pre-sale support tickets
after launch

 **2.4×**

Average order value for
configurator builds vs.
single-item purchases

 **4.9/5**

Post-purchase satisfaction
score for configurator
orders

“

We expected a product page upgrade.
ITForce built us an actual sales system.’

— Game-X founder

GAMEX

✓ All good

Share Build

✓ 1 CPU

✓ 2 Motherboard

✓ 3 RAM

✓ 4 GPU

✓ 5 PSU

6 Case

Your Build Performance

1080p Ultra FPS

185+

Excellent

1440p Ultra FPS

125+

Excellent

Estimated Power Draw

432W

Optimal

Build Confidence

Compatibility

All compatible

✓

Bottlenecks

None detected

✓

Est. Power Headroom

28% remaining

✓

Thermal Check

All good

✓

Est. Power

432W

Total

28 500 ₴

Add to Cart →



Trusted by
5 200+ gamers

Fast delivery
across Ukraine

Official parts
& warranty

Expert support
7 days a week

OpenCart, extended the right way.

A purpose-built architecture inside OpenCart delivers a powerful configurator, real-time compatibility, and a seamless shopping experience.

TECH STACK



OpenCart 3.x



PHP 8.1



MySQL



Vanilla JS

Custom OC
Module API

Git / CI

PROJECT TIMELINE

1

**Weeks 1–2:**Discovery, compatibility matrix design,
DB schema

2

**Weeks 3–4:**

Configurator module + admin UI

3

**Weeks 5–6:**

Compatibility engine + cart integration

4

**Week 7:**

QA, cross-browser, mobile testing

5

**Week 8:**Staging → Production deploy
+ team training